Wireless Tape Measurer

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TECHIN 514
Final Project

Github Repo: https://github.com/vpon-lab/514-Final-Project

UNIVERSITY of WASHINGTON

Measuring alone

Problem to Solve:

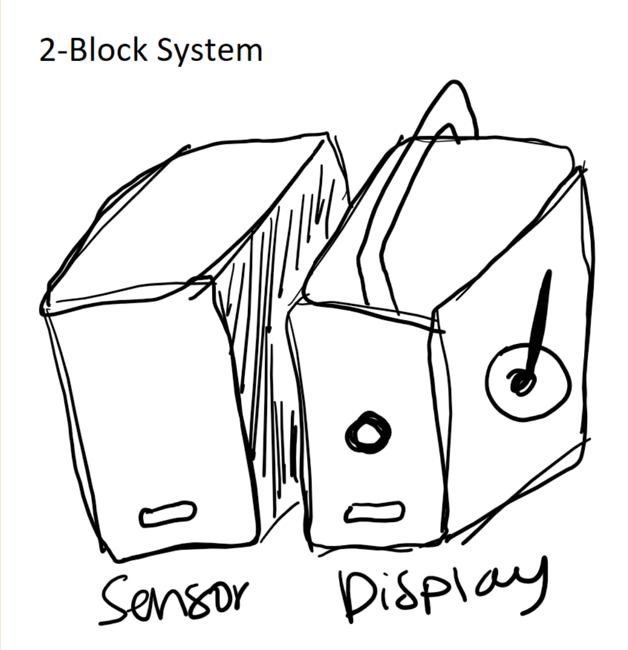
- **Wanting to measure bedroom space on your own with out another person's assistance.**
- **Mitigate sag from a physical tape measurer.**

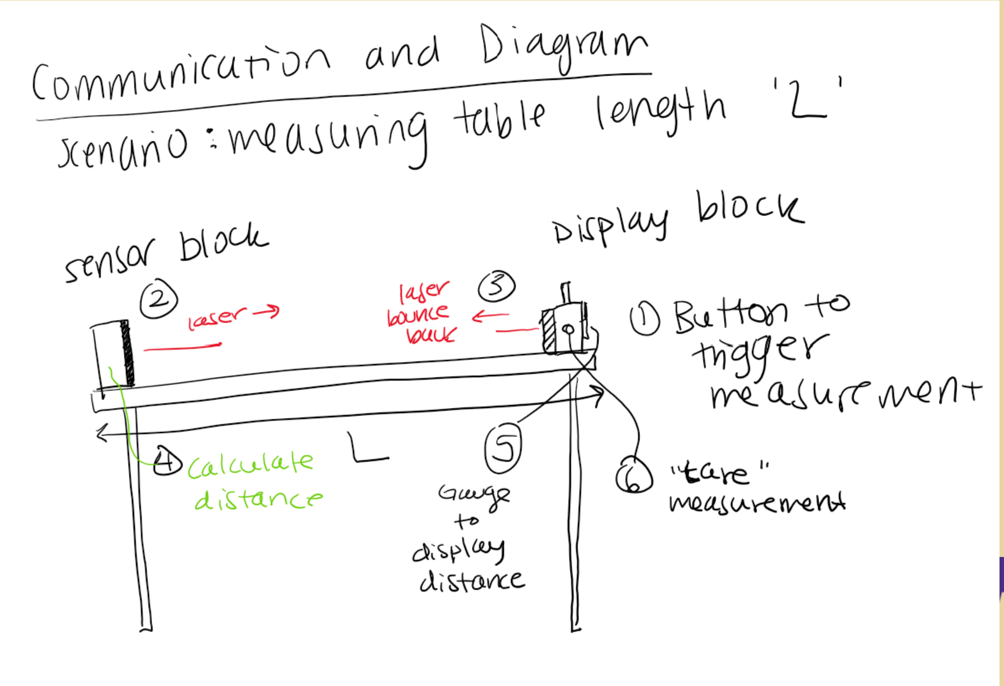
Measuring alone

A proposed Solution:

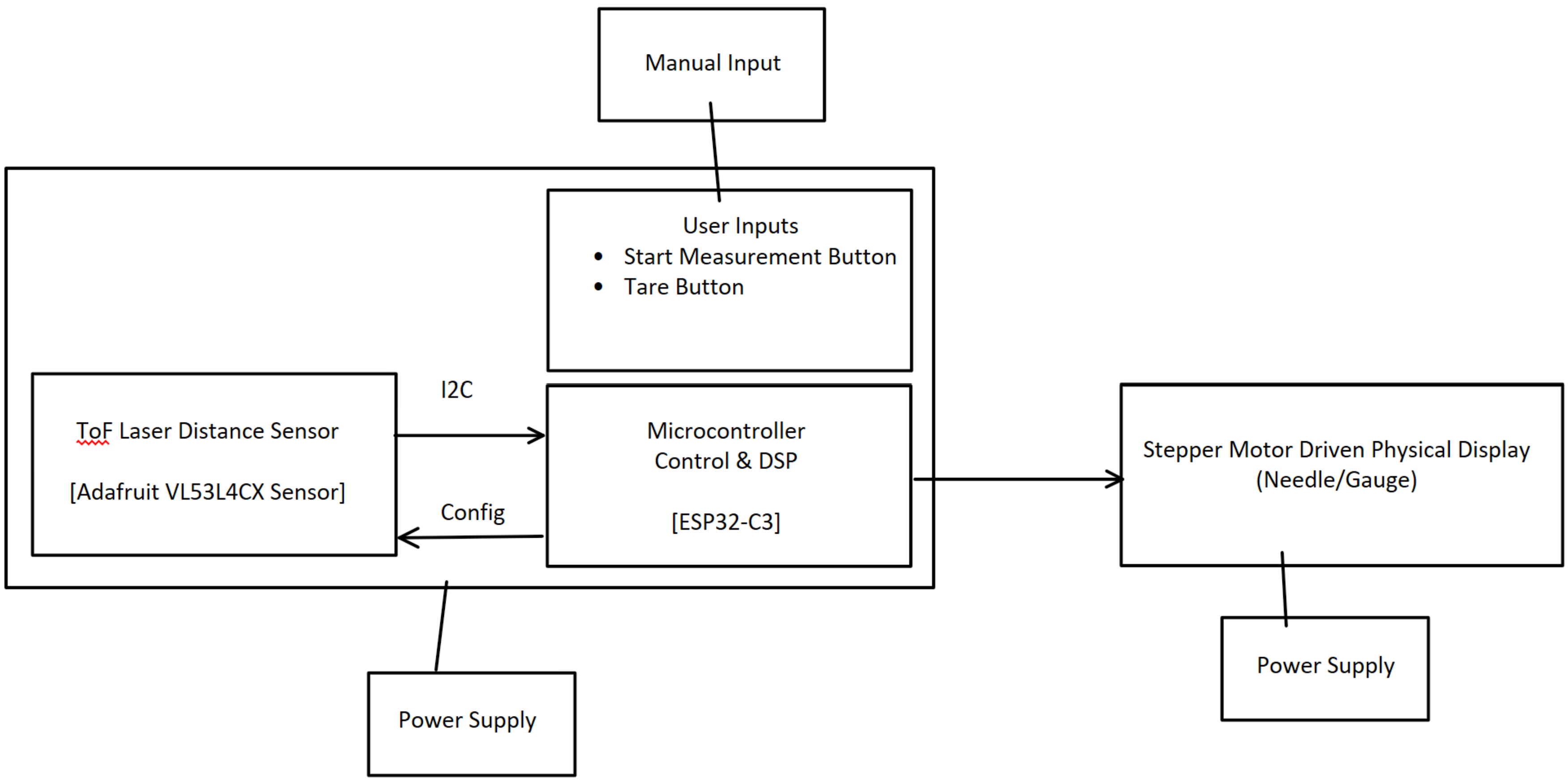
> **Wireless tape measurement**

> **2-piece system to measure from point A to point B**





System Architecture



Sensing + Display Device

Due to multiple ESP32C3 Battery pad mishaps, this is not battery powered for cautionary measures

Sensor Device

XIAO ESP32C3

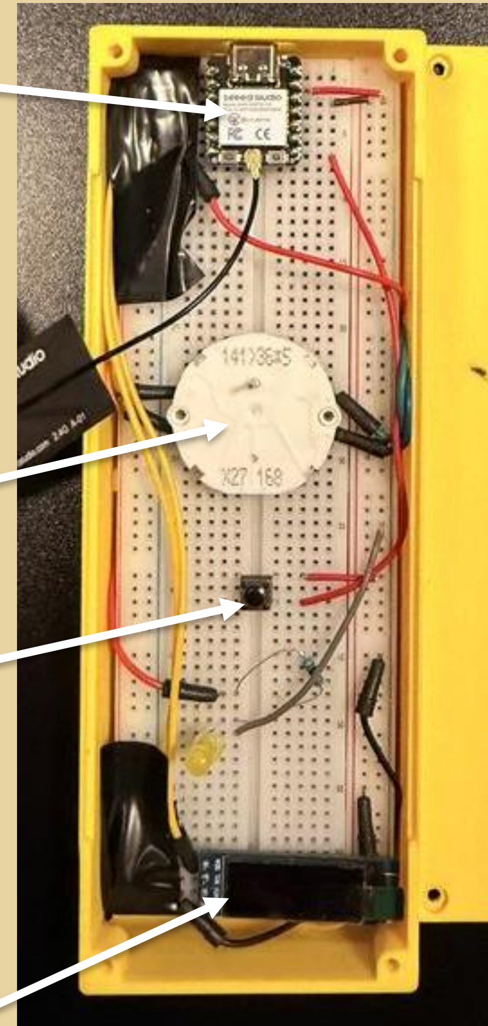
Time of Flight Laser Sensor

X27 Stepper Motor

"Tare" Button

Power On/Off Switch

OLED Display

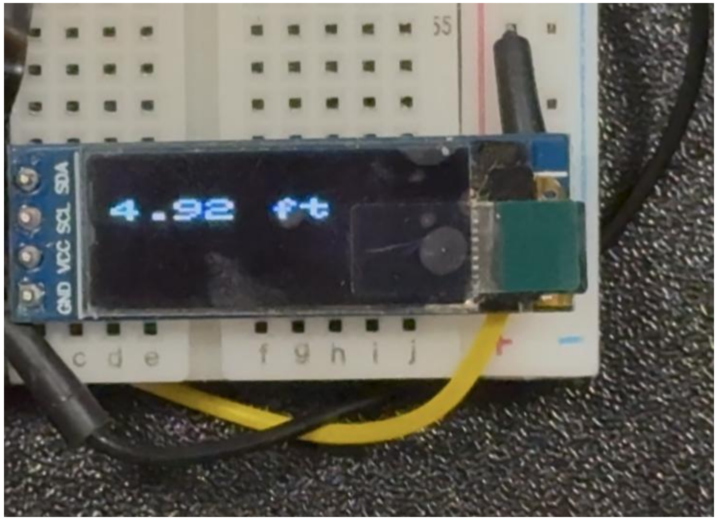


Display Device

Signal Processing

> **Moving average filter, numbers were the same at the first decimal place before filter.**

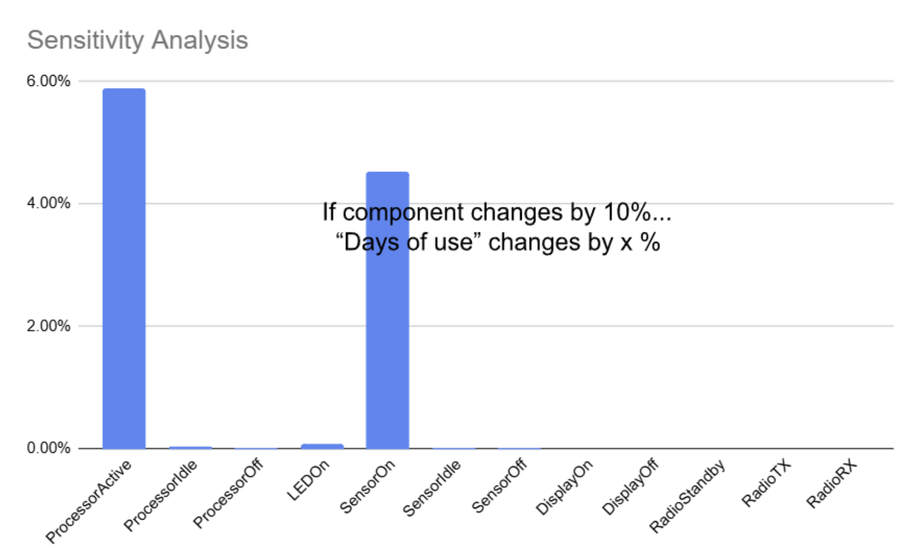
> **Accuracy:**
  - **±0.01 ft**
  - **±3mm**
  - **Degrades with distance**



Display Device shows sensor jumping between measurements – from table to ceiling

Battery Considerations – Sensing device

- 3.7V Lipo or Li-Ion Battery
- Compact easy to design with in enclosure and long lasting
- Based on 2hrs/day of use



Total power in profile (mw)			Maximum Time	
"off"	0.0396	mW	150261.1	hours
"sensing"	79.136475	mW	75.2	hours
"interactive"	80.8236	mW	73.6	hours
Effective Battery Capacity				
	5950.34	mW*h		
Days of Use	**18.55**	**days**		
Hours of Use	**445.28**	**hours**		

Appendix

Anything with "$0" was free in the Maker Space

Budget Summary

Supplies		
Purchased	Qty	Total Price
XIAO ESP32C3	2	0
ST VL53L4CX	2	
3.7V Li-Ion Batteries	2	$14.96
Battery holders	2	$5.99
X27 Stepper Motor	1	0
3.6 kOhm Resistors	2	0
220 Ohm Resistor	1	0
Red LED	1	0
OLED SSD1306	1	0
Tactile Switch	1	0
Single-Pole, Double-Throw (SPDT) slide switch	2	0
3M Reflective Tape	1	$9.05
Plexiglass Clear Acrylic Sheets Panel	1	$9.99
Total		39.99

Future Work

- Sensor Calibration to get rid of cross-talk from clear acrylic window.
- Sensor Calibration to find max capable distance based on Bluetooth connection compatibility
- Develop a more ergonomic and detachable/portable enclosure(s)
- Proper PCB design for easy battery connection to Display device.

Appendix

Power Model

System Parameters (defined by hardware)				**Profiles (usage of each component mode - defined by software and usage)**			
	from the datasheets						
Supply (V)	3.3			"off"	"sensing"	"interactive"	
Processor			**Current Consumption (mA)**				
Active (modeem sleep)	89	mW	27	0%	10%	90%	
Idle	0	mW	0.13	0%	85%	10%	
Sleep	0	mW	0.005	100%	5%	0%	
LED							
On	1	mW	0.17	0%	100%	100%	
Sensor							
On	69	mW	21	0%	100%	0%	
Idle	0	mW	0.009	0%	0%	100%	
Off	0	mW	0.007	100%	0%	0%	
Display							
On	0	mW	0	0%	0%	0%	
Off (leakage)	0	mW	0	0%	0%	0%	
Radio							
Data Rate	400	bps					
Standby Power	3	mW					
TX Power	85	mW					
RX Power	85	mW					
				20	2	2	hours/day typical usage
Battery							
Capacity	1892	mAh					
Nominal Voltage	4	V					
Regulator Efficiency	85%						

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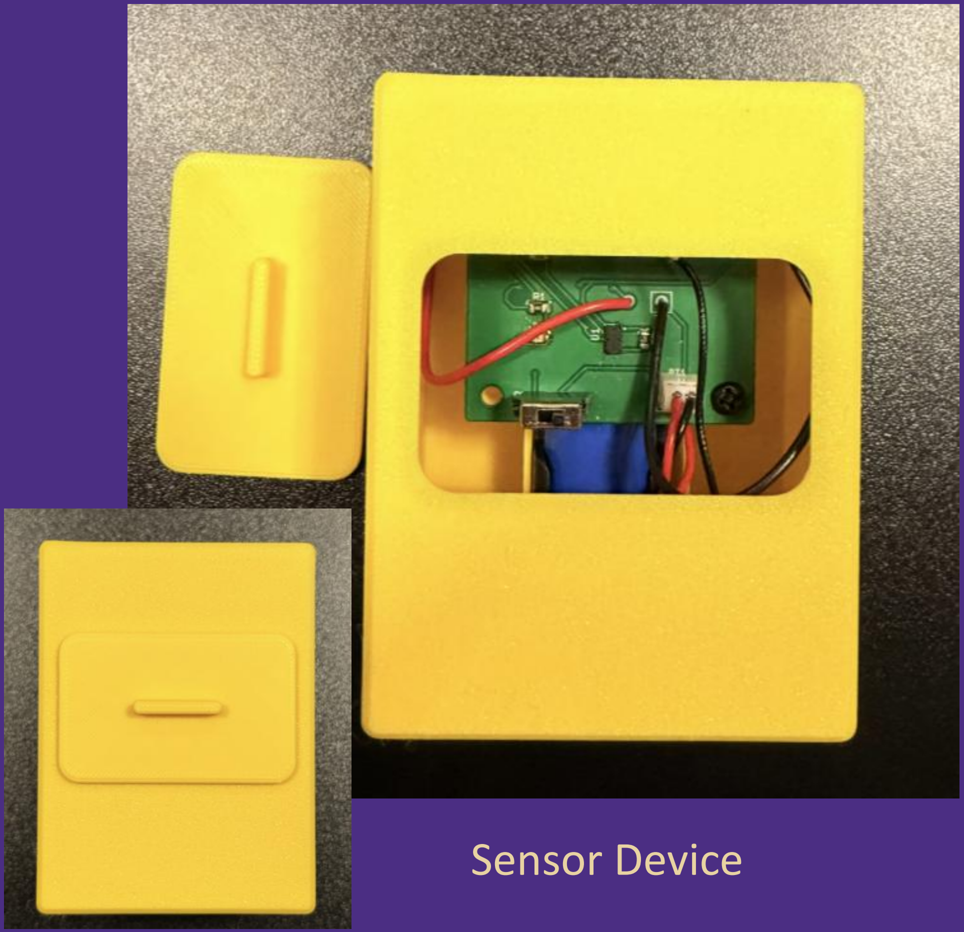
Enclosure Design



Sensor Device

Display Device

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